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## Isogeometric Analysis (IGA) for Components

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## ABSTRACT

Isogeometric Analysis (IGA) promises *time savings* and *increased accuracy* relative to traditional Finite Element Analysis (FEA) approaches.

The savings arise in part from the elimination of the mesh generation step, which is often time consuming and error prone, which we categorize as **analyst time**. The second component of time savings is **compute time**, which arises from faster convergence rates for IGA relative to traditional FEA. For a specific level of accuracy, IGA can often achieve the result with fewer degrees of freedom (DOF) and less computational effort.

This report examines the performance premise of IGA through a series of case studies involving engineering components with complex geometries. We compare IGA results against traditional FEA results for modal and static analyses. The report quantifies both analyst time and compute time savings, while also discussing challenges encountered during the implementation of IGA workflows.

The overarching goal of this work is to raise awareness of IGA technology within the Sandia engineering analysis community.

*To do:*

- *Joe Bishop as reviewer (jebisho@sandia.gov)*

Source document reference: *brief/iga\_for\_components/main.tex*

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## 1. INTRODUCTION

Isogeometric Analysis (IGA) represents a paradigm shift in computational mechanics by unifying the mathematical frameworks of Computer-Aided Design (CAD) and numerical analysis. Introduced by Hughes *et al.* in 2005 [3], IGA employs the same basis functions—typically Non-Uniform Rational B-Splines (NURBS), T-splines, or subdivision surfaces—for both the geometric representation and the solution approximation.

Traditional Finite Element Analysis (FEA) has served as the cornerstone of computational engineering for decades, offering robust capabilities for solving partial differential equations across diverse applications. Its strengths include mature software ecosystems, well-established theoretical foundations, extensive validation in industrial settings, and flexibility in handling complex physics through various element formulations. However, FEA suffers from one large inherent limitation: The CAD geometry must be meshed, which only approximates the CAD description. The mesh generation process introduces geometric approximation errors, requires significant preprocessing time, and creates  $C^0$ -continuous approximations that may be inadequate for higher-order problems such as thin shell analysis, phase-field modeling, and gradient-based theories. Additionally, design optimization workflows necessitate repeated geometry-mesh conversions, introducing inefficiencies and potential information loss.

IGA addresses these deficiencies through its geometric fidelity and mathematical elegance. By maintaining exact geometric representation at all refinement levels, IGA eliminates geometric discretization errors and the cumbersome mesh generation step. The inherent higher-order continuity of spline basis functions—providing  $C^1$ ,  $C^2$ , or arbitrary  $C^k$  continuity—yields superior accuracy per degree of freedom for many problem classes, particularly those involving fourth-order operators or requiring smooth solution fields. The seamless integration with CAD systems enables direct analysis on design geometries and facilitates efficient shape optimization. Furthermore, IGA demonstrates enhanced performance in capturing complex phenomena such as boundary layers, stress concentrations, and wave propagation with coarser discretizations than traditional FEA.

Despite its advantages, IGA faces several challenges that have tempered its widespread industrial adoption. The technology requires specialized software implementations with limited commercial availability compared to mature FEA platforms. Integration of trimmed surfaces, boolean operations, and complex multi-patch geometries from industrial CAD models remains an active research area. Volumetric parameterization of three-dimensional solids presents non-trivial difficulties, particularly for complex topologies. The computational cost per degree of freedom can be higher due to increased basis function support and quadrature requirements. Additionally, the engineering community's familiarity with traditional FEA methods, coupled with substantial investments in existing workflows and validation databases, creates institutional inertia against adoption.

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## 2. BACKGROUND

We review progress of IGA technology from perspectives of industry, academia, and government laboratories.

Within industry, several commercial software packages now support IGA capabilities. Notable examples include Ansys, Siemens NX Nastran, Dassault Systèmes Abaqus, Altair HyperWorks, and Coreform Flex. These packages offer varying levels of IGA functionality, from basic NURBS-based elements to more advanced spline technologies and multi-patch coupling methods.

Ansys (via LS-DYNA) is currently the most prominent large-scale commercial software vendor of IGA. Since the acquisition of LSTC, Ansys has integrated IGA capabilities into LS-DYNA, making it the most mature commercial solver for IGA applications.

LS-DYNA supports NURBS-based shells and solids, with specific emphasis on crash analysis and large-deformation problems. The first NURBS-based shell formulations appeared in LS-DYNA in 2011. Thereafter, IGA capabilities were expanded to solid elements, implicit and explicit analysis, contain, trimmed NURBS shells, and unstructured boundary-fitted approaches using Bézier extraction. In late 2025 and early 2026, Ansys expanded its “Trimmed IGA Solids” capabilities, which allows for immersed boundary analysis wherein the physics is solved on a background grid that respects the CAD geometry via trimming operations. No traditional finite element mesh is required [4].

One example of IGA performance on thin-walled structures with complex surface geometry was provided by Novozhilov and Leidinger [4]. Their use of the trimmed IGA approach permitted larger time steps and required fewer elements to achieve similar accuracy compared to traditional FEA approaches. Specially, their IGA model delivered comparable accuracy to a TET-16 formulation, while reducing compute time by a factor of  $21\times$ .

Siemens Digital Industries Software (NX and Simcenter Nastran) has also made significant strides in IGA technology. Siemens offers NURBS-based shell and solid elements, with capabilities for multi-patch coupling and adaptive refinement. Their IGA implementations focus on structural analysis, including static, dynamic, and thermal problems. As of June 2025 and continuing into 2026, Siemens moved beyond previews of NX Performance Predictor. While not branded as IGA, the underlying technology leverages spline-based representations to enhance accuracy and efficiency in performance prediction workflows. Siemens also has integrated subdivision modeling (an alternative to NURBS) that maintains higher-order ( $C^2$ ) [5].

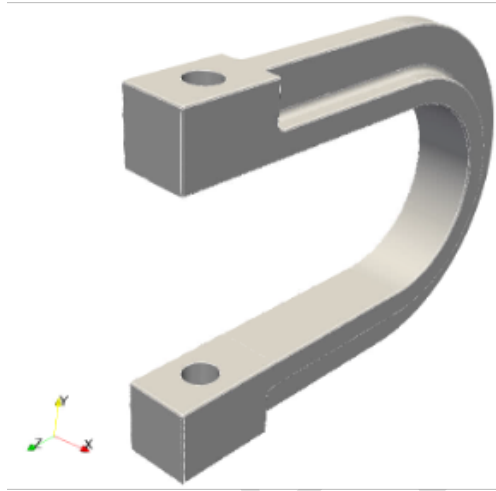
Dassault Systèmes Abaqus has been historically more conservative with native IGA integration. Abaqus supports IGA through high-fidelity toolkits and user-defined elements (UEL).

Altair HyperWorks has explored IGA capabilities through research collaborations and prototype implementations, but has yet to fully integrate IGA into its mainstream offerings. The 2026 release of HyperWorks included IGA-like capabilities through its “warp-map” tools [1].

Coreform is a computational mechanics software company that specializes in isogeometric analysis (IGA) and advanced meshing technologies. Coreform’s flagship product is Coreform Cubit, a powerful meshing and geometry processing tool that supports IGA workflows. Coreform also offers Coreform Flex, a solver specifically designed for isogeometric analysis applications. Michael Scott is the CEO of Coreform, Matt Sederberg is the Chief Strategic Officer, Greg Vernon is the Director, and Thomas J.R. Hughes is the Senior Advisor [2]. The authors of this SAND report worked periodically with Sederberg and Vernon during Q1 of FY2026 to obtain access to Coreform Flex and to discuss IGA capabilities and workflows.

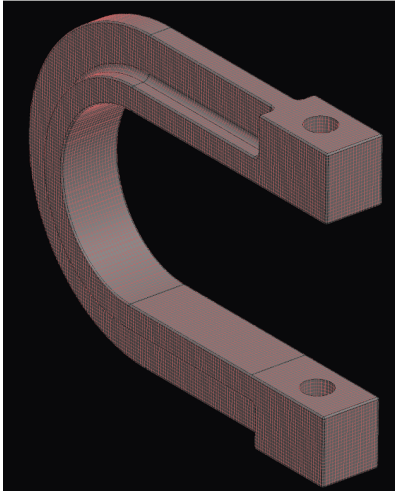
### 3. MODELS

#### 3.1. C-Frame

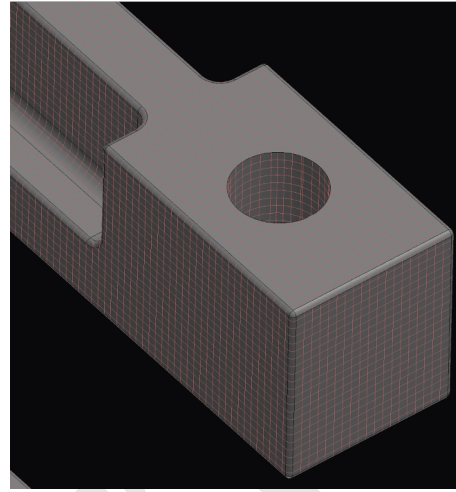


**Figure 3-1. C-frame geometry.**

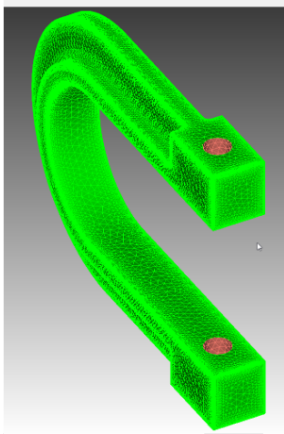
- Analysis type: modal
- Material: isotropic linear elastic
- Mesh: IGA
- Geometry: C-frame
- Boundary conditions: fixed at base
- Loads: N/A
- Solver: direct
- Geometry included in Coreform Flex distribution, no additional cleanup needed for IGA.
- Modal analysis performed
- Requested modes in (1, 4000) Hz
- Solved locally on laptop 1 cpu, Windows 11, 32 GB RAM, Intel i7 (13th gen)



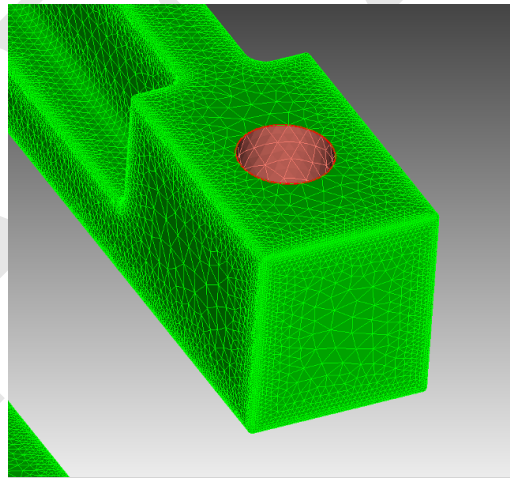
(a) IGA model with 383,250 DOF.



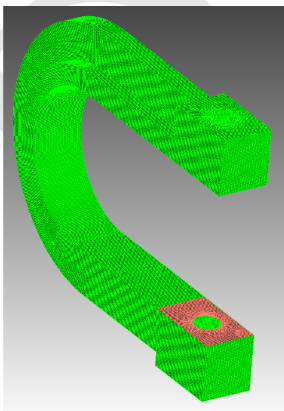
(b) IGA model zoomed in.



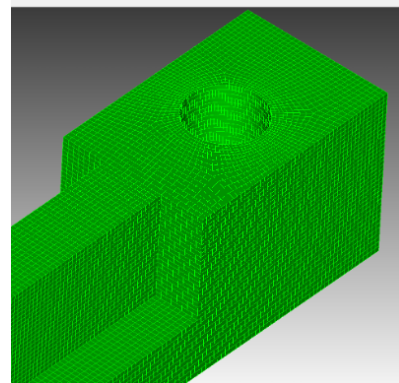
(c) Tetrahedral FEA model with 4,072,959 DOF.



(d) Tetrahedral FEA model zoomed in.



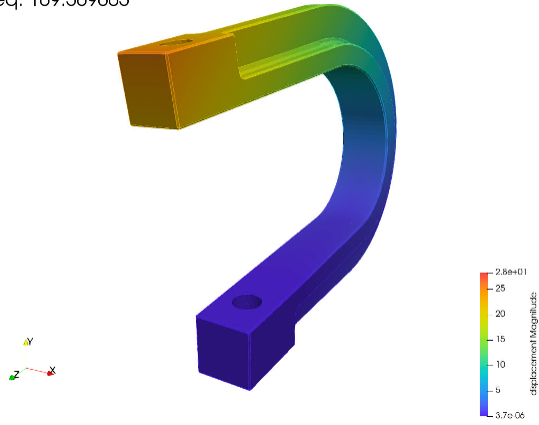
(e) Hexahedral FEA model with 1,167,897 DOF.



(f) hexahedral FEA model zoomed in.

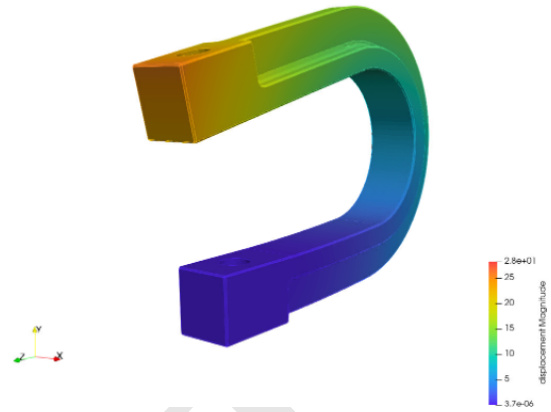
Figure 3-2. Fine mesh models for C-frame geometry. (a,b) IGA model. (c,d) Tetrahedral FEA model. (e,f) Hexahedral FEA model.

Freq: 169.569605



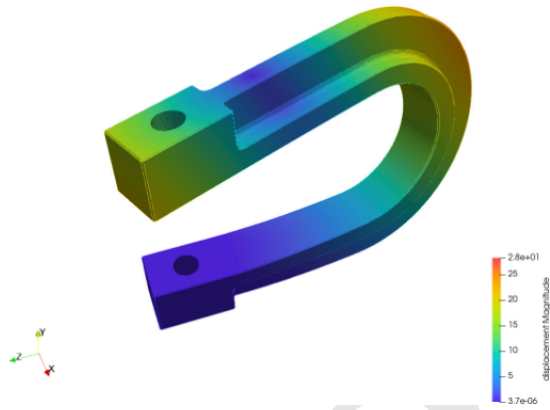
(a) Mode 1

Freq: 301.280628



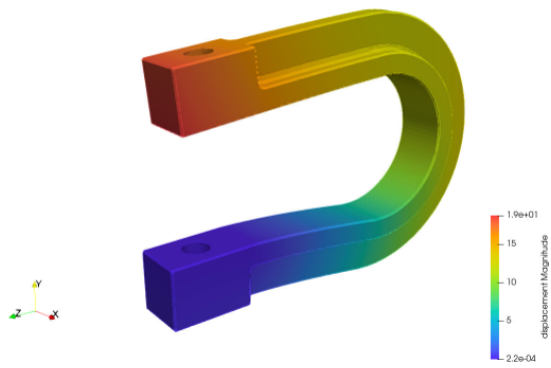
(b) Mode 2

Freq: 409.101071



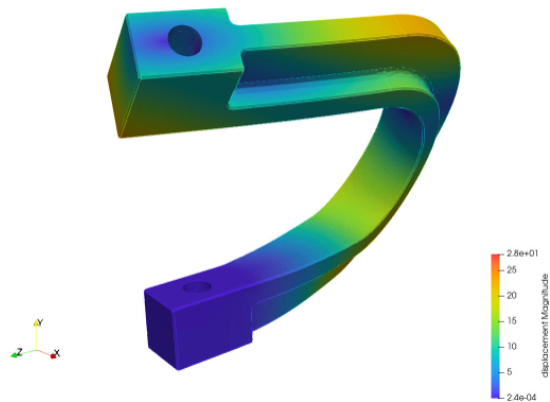
(c) Mode 3

Freq: 536.189570



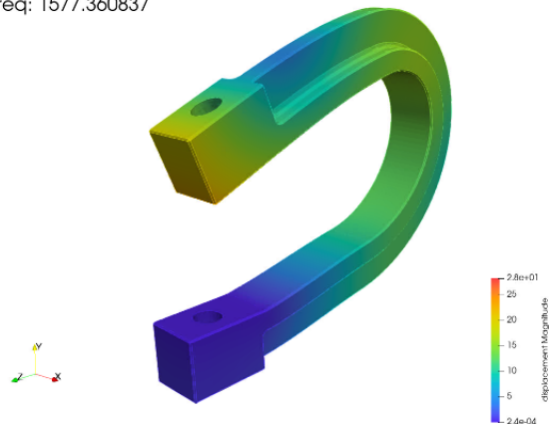
(d) Mode 4

Freq: 1221.273218



(e) Mode 5

Freq: 1577.360837

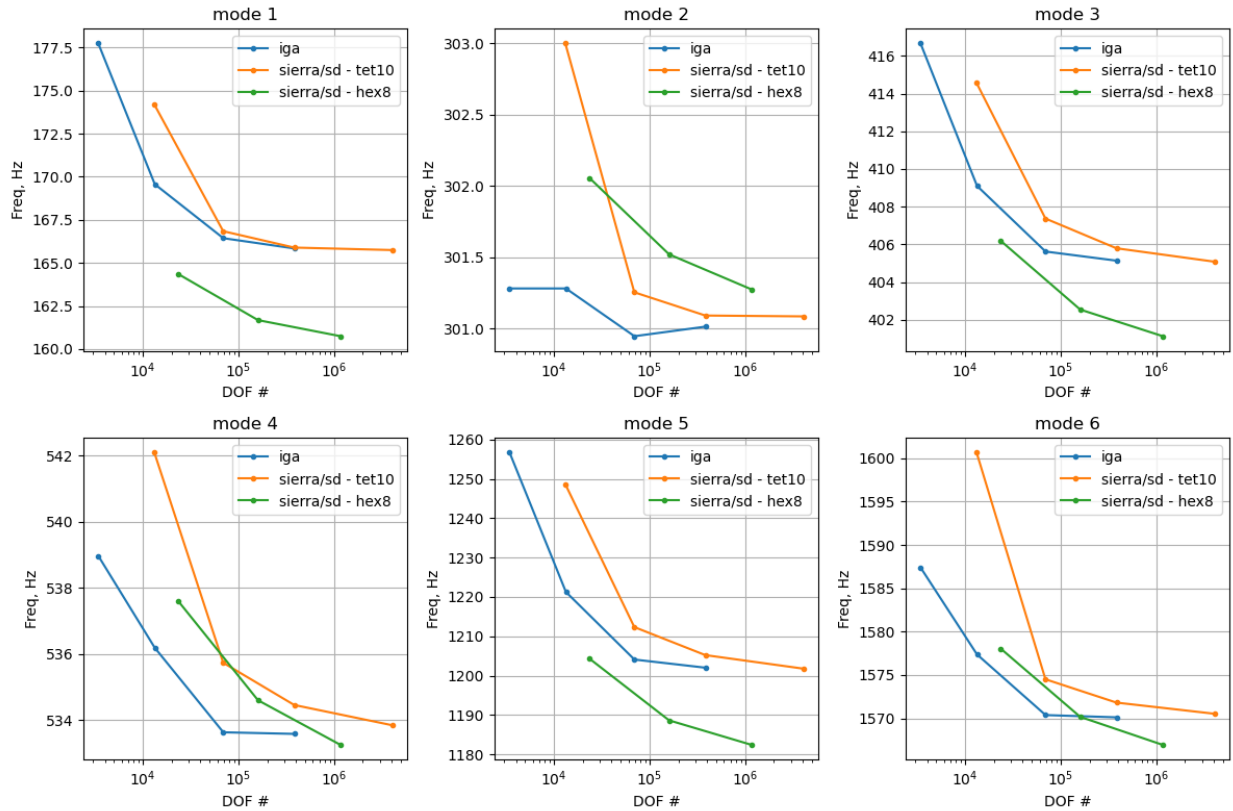


(f) Mode 6

Figure 3-3. Mode shapes and frequencies for the C-frame IGA model. The first six modes are shown. The IGA model used is first refinement model (13,332 DOF).

Status: Alejandro trying to match the DOF x-axis. David to set up meeting with Alejandro and Chad to discuss results. Alejandro has DOF data for existing FEA models. Alejandro defeatured hex model, but not the tet model. David to email to Chad four IGA models (coarse, fine, finer, finest) by next Friday 19 Dec 2025. David to put python, Excel, and supporting files on the repository.

DRAFT

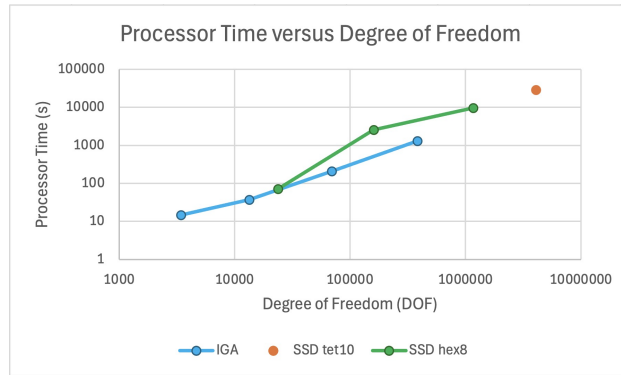


**Figure 3-4. Model frequency convergence as a function of degree of freedom (DOF) for IGA and FEA models. FEA included SSD tetrahedral (10-noded) and SSD hexahedral (8-noded) meshes. DOF count increases from left (coarse) to right (fine). The IGA model shows faster convergence with DOF compared to the traditional FEA models. DOF count specifics appear in Table 3-1.**

**Table 3-1. C-frame model resolutions (DOF #), run times, and number of processors.**

| Analysis Type | Coarse        | ← ..... →       |                 |                  | Fine |
|---------------|---------------|-----------------|-----------------|------------------|------|
| IGA           | 3,420         | 13,332          | 69,426          | 383,250          |      |
|               | 15s (1 core)  | 37s (1 core)    | 209s (1 core)   | 1,303s (1 core)  |      |
| SSD tet10     | 13,146        | 69,234          | 382,791         | 4,072,959        |      |
|               | r3c2          | r3c3            | r3c4            | 225s (128 cores) |      |
| SSD hex8      | 23,682        | 160,059         | 1,167,897       | —                |      |
|               | 6s (12 cores) | 20s (128 cores) | 75s (128 cores) | —                |      |

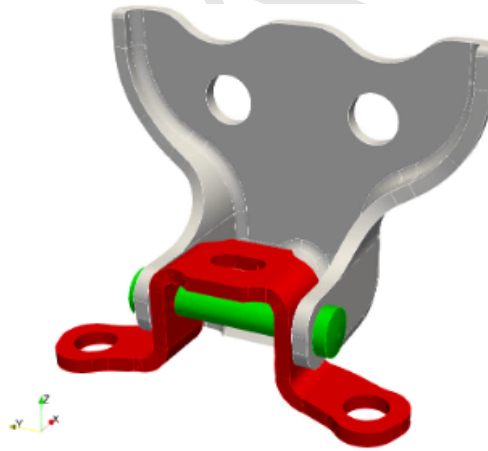
The data in Table 3-1 summarizes the model resolutions (DOF counts), run times, and number of processors used for each C-frame model. The IGA models were run on a single core, while the FEA models were run on multiple cores to reduce wall time. The cores were multiplied by wall time to estimate total processor time for the FEA models. These comparisons are shown in Figure 3-5. The source for Figure 3-5 is the Excel file *cframe\_modal/C-frame\_processor\_time\_vs\_dof.xlsx* in the *cee-gitlab.sandia.gov/structMechTools/iga* repository.



**Figure 3-5. Processor time versus degree of freedom (DOF) for C-frame IGA and FEA models. The IGA model shows lower processor time for a given DOF count compared to the traditional FEA models.**

### 3.2. Door Hinge

- Analysis type: nonlinear statics
- Features: frictional contact, plasticity



**Figure 3-6. Door hinge geometry used for nonlinear static analysis.**

### 3.3. Circuit Board

- Analysis type: nonlinear statics, modal
- Features: linear elastic materials
- Note: required slight geometry clean up

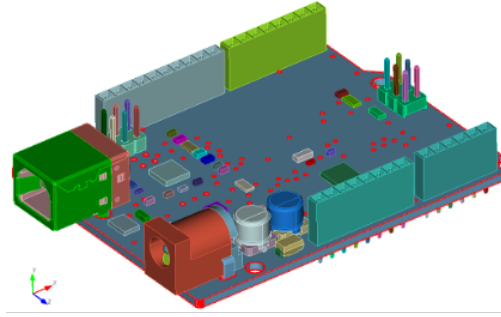


Figure 3-7. Circuit board geometry used for nonlinear static and modal analysis.

#### 3.4. UUR Discriminator

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#### **4. METHODS**

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## **5. RESULTS**

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## **6. CONCLUSION**

That is, as they say, all, folks.

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## REFERENCES

- [1] Altair Engineering Inc. (2026). *HyperWorks 2026: Design and Simulation for the Future*. <https://altair.com/hyperworks-2026>. (Accessed: Jan. 16, 2026).
- [2] Coreform LLC. (2026). "About Coreform." <https://coreform.com/about/>. (Accessed: Jan. 16, 2026).
- [3] Hughes, T.J.R., Cottrell, J.A., and Bazilevs, Y. (2005). "Isogeometric analysis: CAD, finite elements, NURBS, exact geometry and mesh refinement." *Computer Methods in Applied Mechanics and Engineering*. <https://doi.org/10.1016/j.cma.2004.10.008>
- [4] Novozhilov Y. and Leidinger L. (2025). "Trimmed IGA Solids in LS-DYNA: CADFEM Findings, 2025 Ansys EMEA Transportation Summit and LS-DYNA Conference, Munich, Germany, 2025.
- [5] Siemens Digital Industries Software. (2025). *What's New In NX June 2025*. <https://plm.sw.siemens.com/en-US/nx/cad-online/latest-nx-cad-version/>. (Accessed: Jan. 16, 2026).

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## APPENDIX A. Appendices

### A.1. C-Frame

#### A.1.1. jobs\_coarse.iga

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4   "data_output_interval": "intervals_1",
5   "flex_models": [
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7       "database_name": "geom",
8       "label": "flex_models_1",
9       "parts": [
10        {
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12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
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18            "optimization_residual_tolerance": 1.0000000000000002e-12,
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20            "quadrature": "QP1"
21          }
22        }
23      ],
24      "small_cell_volume_ratio": 0
25    }
26  ],
27  "functions": [
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31      },
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33    }
34  ],
35  "intervals": [
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39      "step_increment": 1,
40      "stop_step": 20,
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42    }
43  ],
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60          "boundary_conditions_1"
61        ],
62        "flex_model": "flex_models_1",
63        "modal_solution_method": "modal_solution_methods_1",
64        "outputs": [
65          "outputs_1"
66        ]
67      }
68    }
69  ]
70 }
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68 }
69 ],
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76         "z"
77       ],
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81         "value": 10
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87         "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88         "quadrature": "QP1"
89       },
90       "set": "fixed"
91     }
92   ],
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96         "maximum_frequency": 4000,
97         "minimum_frequency": 1
98       },
99       "label": "modal_solution_methods_1"
100     }
101   ],
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111           "displacement": [
112             "all"
113           ]
114         }
115       },
116       "label": "outputs_1",
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118     }
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120 }
121 }

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### A.1.2. jobs\_fine.iga

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9       "parts": [
10        {
11          "label": "parts_1",
12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
15          "quadrature_options": {
16            "optimization_lawson_hanson_solve_completion": true,
17            "optimization_method": "tet_pruning",
18            "optimization_residual_tolerance": 1.0000000000000002e-12,
19            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20            "quadrature": "QP1"
21          }
22        }
23      ],
24      "small_cell_volume_ratio": 0
25    }
26  ],
27  "functions": [

```

```

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30       "value": 1
31     },
32     "label": "functions_1"
33   }
34 ],
35 "intervals": [
36   {
37     "label": "intervals_1",
38     "start_step": 0,
39     "step_increment": 1,
40     "stop_step": 20,
41     "time_increment": 1
42   }
43 ],
44 "label": "iga_1",
45 "materials": [
46   {
47     "elastic": {
48       "poissons_ratio": 0.29,
49       "youngs_modulus": 28000000
50     },
51     "label": "materials_1",
52     "mass_density": 0.000735
53   }
54 ],
55 "procedures": [
56   {
57     "label": "procedures_1",
58     "structural_modal_analysis": {
59       "boundary_conditions": [
60         "boundary_conditions_1"
61       ],
62       "flex_model": "flex_models_1",
63       "modal_solution_method": "modal_solution_methods_1",
64       "outputs": [
65         "outputs_1"
66       ]
67     }
68   }
69 ],
70 "structural_modal_analysis_definitions": {
71   "boundary_conditions": [
72     {
73       "components": [
74         "x",
75         "y",
76         "z"
77       ],
78       "label": "boundary_conditions_1",
79       "penalty": {
80         "dimensionless": true,
81         "value": 10
82       },
83       "quadrature_options": {
84         "optimization_lawson_hanson_solve_completion": true,
85         "optimization_method": "tet_pruning",
86         "optimization_residual_tolerance": 1.0000000000000002e-12,
87         "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88         "quadrature": "QP1"
89       },
90       "set": "fixed"
91     }
92 ],
93 "modal_solution_methods": [
94   {
95     "frequency_extraction": {
96       "maximum_frequency": 4000,
97       "minimum_frequency": 1
98     },
99     "label": "modal_solution_methods_1"
100   }
101 ],
102 "outputs": [
103   {
104     "field": {
105       "database_name": "cframe_modal_fine",
106       "external_surfaces_only": false,
107       "include_integration_point_output": false,
108       "interval": "intervals_1",
109       "part": "part_1",
110       "variables": {
111         "displacement": [
112           "all"
113         ]
114       }
115     },
116     "label": "outputs_1",
117     "tessellate": true
118   }
119 ]

```

```
120 }
121 }
```

### A.1.3. jobs\_fine\_x2.iga

```
1 {
2   "coreform_iga_version": "2025.7",
3   "data_output_file": "cf_iga_data_output.json",
4   "data_output_interval": "intervals_1",
5   "flex_models": [
6     {
7       "database_name": "geom",
8       "label": "flex_models_1",
9       "parts": [
10        {
11          "label": "parts_1",
12          "material": "materials_1",
13          "material_model": "elastic",
14          "part": "part_1",
15          "quadrature_options": {
16            "optimization_lawson_hanson_solve_completion": true,
17            "optimization_method": "tet_pruning",
18            "optimization_residual_tolerance": 1.0000000000000002e-12,
19            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
20            "quadrature": "QP1"
21          }
22        }
23      ],
24      "small_cell_volume_ratio": 0
25    }
26  ],
27  "functions": [
28    {
29      "constant": {
30        "value": 1
31      },
32      "label": "functions_1"
33    }
34  ],
35  "intervals": [
36    {
37      "label": "intervals_1",
38      "start_step": 0,
39      "step_increment": 1,
40      "stop_step": 20,
41      "time_increment": 1
42    }
43  ],
44  "label": "iga_1",
45  "materials": [
46    {
47      "elastic": {
48        "poissons_ratio": 0.29,
49        "youngs_modulus": 28000000
50      },
51      "label": "materials_1",
52      "mass_density": 0.000735
53    }
54  ],
55  "procedures": [
56    {
57      "label": "procedures_1",
58      "structural_modal_analysis": {
59        "boundary_conditions": [
60          "boundary_conditions_1"
61        ],
62        "flex_model": "flex_models_1",
63        "modal_solution_method": "modal_solution_methods_1",
64        "outputs": [
65          "outputs_1"
66        ]
67      }
68    }
69  ],
70  "structural_modal_analysis_definitions": {
71    "boundary_conditions": [
72      {
73        "components": [
74          "x",
75          "y",
76          "z"
77        ],
78        "label": "boundary_conditions_1",
79        "penalty": {
```

```

80     "dimensionless": true,
81     "value": 10
82 },
83 "quadrature_options": {
84     "optimization_lawson_hanson_solve_completion": true,
85     "optimization_method": "tet_pruning",
86     "optimization_residual_tolerance": 1.0000000000000002e-12,
87     "optimized_cut_quadrature_strategy": "reduce_from_uncut",
88     "quadrature": "QP1"
89 },
90 "set": "fixed"
91 }
92 ],
93 "modal_solution_methods": [
94     {
95         "frequency_extraction": {
96             "maximum_frequency": 4000,
97             "minimum_frequency": 1
98         },
99         "label": "modal_solution_methods_1"
100     }
101 ],
102 "outputs": [
103     {
104         "field": {
105             "database_name": "cframe_modal_fine_x2",
106             "external_surfaces_only": false,
107             "include_integration_point_output": false,
108             "interval": "intervals_1",
109             "part": "part_1",
110             "variables": {
111                 "displacement": [
112                     "all"
113                 ]
114             }
115         },
116         "label": "outputs_1",
117         "tessellate": true
118     }
119 ]
120 }
121 }

```

## A.2. Door Hinge

### A.2.1. door\_hinge\_failure\_workflow.txt

```

1  #=Door Lock Hinge Assembly Problem
2  reset
3  dev rimport "/tutorials/door_lock_assembly/door_lock_assembly.sat"
4  #!=Door Lock Hinge Assembly Problem
5  #=DEFINE PARTS AND SETS
6  name part 1 "door_link"
7  name part 2 "door_hinge"
8  name part 3 "door_pin"
9  set "door_link_bottom_surfaces" add surfaces [21 29]
10 set "door_hinge_bolt_surfaces" add surfaces [111 115]
11 #!=DEFINE PARTS AND SETS
12 #=MESH GENERATION
13 mesh "mesh_door_link" rectilinear degree 2 continuity 1 element_size [1 1 1] padding [2 2 2]
14 part "door_link" mesh "mesh_door_link"
15 part "door_link" volume_box
16 mesh "mesh_door_hinge" rectilinear degree 2 continuity 1 element_size [1 1 1] padding [2 2 2]
17 part "door_hinge" mesh "mesh_door_hinge"
18 part "door_hinge" volume_box
19 mesh "mesh_door_pin" rectilinear degree 2 continuity 1 element_size [1 1 1] padding [2 2 2]
20 part "door_pin" mesh "mesh_door_pin"
21 part "door_pin" volume_box
22 #!=MESH GENERATION
23 #=FUNCTIONS
24 functions "aluminum_plastic_hardening" new
25 functions "aluminum_plastic_hardening" piecewise_linear abscissa [0 10]
26 functions "aluminum_plastic_hardening" piecewise_linear ordinate [0 100]
27 functions "steel_plastic_hardening" new
28 functions "steel_plastic_hardening" piecewise_linear abscissa [0 10]
29 functions "steel_plastic_hardening" piecewise_linear ordinate [0 100]
30 functions "constant_1" new
31 functions "constant_1" constant value 1
32 functions "linear_ramp" new
33 functions "linear_ramp" piecewise_linear abscissa [0 1]
34 functions "linear_ramp" piecewise_linear ordinate [0 1]

```

```

35 #!=FUNCTIONS
36 #=MATERIALS
37 materials "aluminum" new
38 materials "aluminum" mass_density 2.7e-9
39 materials "aluminum" neohookean_isotropic_plasticity neohookean youngs_modulus 73.8e3
40 materials "aluminum" neohookean_isotropic_plasticity neohookean poissons_ratio 0.3
41 materials "aluminum" neohookean_isotropic_plasticity isotropic_plasticity yield_stress 322
42 materials "aluminum" neohookean_isotropic_plasticity isotropic_plasticity hardening_function "aluminum_plastic_hardening"
43 materials "steel" new
44 materials "steel" mass_density 2.7e-9
45 materials "steel" neohookean_isotropic_plasticity neohookean youngs_modulus 200e3
46 materials "steel" neohookean_isotropic_plasticity neohookean poissons_ratio 0.29
47 materials "steel" neohookean_isotropic_plasticity isotropic_plasticity yield_stress 2e3
48 materials "steel" neohookean_isotropic_plasticity isotropic_plasticity hardening_function "steel_plastic_hardening"
49 tessellation_options "tessellation_options_1" meshgems average_triangle_edge_subdivisions 4
50 #!=MATERIALS
51 #=FLEX MODELS
52 flex_models "flex_models_1" new
53 flex_models "flex_models_1" database_name "geom"
54 flex_models "flex_models_1" small_cell_volume_ratio 0
55 flex_models "flex_models_1" parts "door_hinge_instance" part "door_hinge"
56 flex_models "flex_models_1" parts "door_hinge_instance" material "aluminum"
57 flex_models "flex_models_1" parts "door_hinge_instance" material_model neohookean_isotropic_plasticity
58 flex_models "flex_models_1" parts "door_hinge_instance" quadrature_options tessellation_option "tessellation_options_1"
59 flex_models "flex_models_1" parts "door_link_instance" part "door_link"
60 flex_models "flex_models_1" parts "door_link_instance" material "aluminum"
61 flex_models "flex_models_1" parts "door_link_instance" material_model neohookean_isotropic_plasticity
62 flex_models "flex_models_1" parts "door_link_instance" quadrature_options tessellation_option "tessellation_options_1"
63 flex_models "flex_models_1" parts "door_pin_instance" part "door_pin"
64 flex_models "flex_models_1" parts "door_pin_instance" material "steel"
65 flex_models "flex_models_1" parts "door_pin_instance" material_model neohookean_isotropic_plasticity
66 flex_models "flex_models_1" parts "door_pin_instance" quadrature_options tessellation_option "tessellation_options_1"
67 #!=FLEX MODELS
68 #=INTERVALS
69 intervals "simulation_interval" new
70 intervals "simulation_interval" start_time 0
71 intervals "simulation_interval" stop_time 1
72 intervals "simulation_interval" time_increment 5e-4
73 intervals "simulation_output_interval" new
74 intervals "simulation_output_interval" use_start_stop_from_interval "simulation_interval"
75 intervals "simulation_output_interval" step_increment 1
76 #!=INTERVALS
77 #=LINEAR EQUATION SOLVERS
78 linear_equation_solvers "direct_lu" new
79 linear_equation_solvers "direct_lu" direct lu
80 linear_equation_solvers "direct_multi_frontal" new
81 linear_equation_solvers "direct_multi_frontal" direct multi_frontal
82 #!=LINEAR EQUATION SOLVERS
83 #=NONLINEAR EQUATION SOLVERS
84 nonlinear_equation_solvers "newton_raphson" new
85 nonlinear_equation_solvers "newton_raphson" newton linear_equation_solver "direct_multi_frontal"
86 nonlinear_equation_solvers "newton_raphson" newton maximum_iterations 20
87 nonlinear_equation_solvers "newton_raphson" newton target_relative_residual 1e-8
88 #!=NONLINEAR EQUATION SOLVERS
89 #=TIMESTEPPERS
90 time_steppers "implicit_dynamics_quasistatic" new
91 time_steppers "implicit_dynamics_quasistatic" generalized_alpha spectral_radius 0
92 time_steppers "implicit_dynamics_quasistatic" generalized_alpha predictor_constant_velocity
93 time_steppers "implicit_dynamics_quasistatic" generalized_alpha nonlinear_equation_solver "newton_raphson"
94 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity maximum_time_step 0.1
95 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity minimum_time_step 1e-5
96 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity decrease_factor 0.25
97 time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity increase_factor 1.5
98 #!=TIMESTEPPERS
99 #=BOUNDARY CONDITIONS
100 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" new
101 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" set "door_link_bottom_surfaces"
102 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement_function "constant_1"
103 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement_components 0 x
104 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement_components 1 y
105 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement_components 2 z
106 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" displacement_scale_factor 0
107 solid_mechanics_definitions boundary_conditions "hold_door_link_bottom" quadrature_options tessellation_option "tessellation_options_1"
108 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" new
109 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" set "door_hinge_bolt_surfaces"
110 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement_function "linear_ramp"
111 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement_components 0 x
112 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement_components 1 y
113 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement_components 2 z
114 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" displacement_scale_factor [0 0 9]
115 solid_mechanics_definitions boundary_conditions "pull_door_hinge_bolts" quadrature_options tessellation_option "tessellation_options_1"
116 #!=BOUNDARY CONDITIONS
117 #=INTERACTIONS
118 solid_mechanics_definitions interaction_properties "friction" new
119 solid_mechanics_definitions interaction_properties "friction" coulomb_friction coefficient_with_regularization friction_coefficient 0.3
120 solid_mechanics_definitions interaction_properties "friction" coulomb_friction coefficient_with_regularization regularization_velocity 0.1
121 solid_mechanics_definitions interactions "frictional_contact" new
122 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact base_contact complement true
123 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact interaction_properties "friction"
124 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact approximate_tangent_iterations 6
125 solid_mechanics_definitions interactions "frictional_contact" quadrature_options tessellation_option "tessellation_options_1"
126 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact activation_distance value 0.5

```

```

127 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact activation_zone_influential_radius_factor 1.0
128 solid_mechanics_definitions interactions "frictional_contact" mechanical_contact potential_cutoff_distance_factor 0.5
129 #!=INTERACTIONS
130 #=-PROBES
131 solid_mechanics_definitions probes "door_link_reaction_force" new
132 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity
133 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity use_set_from_boundary_condition "hold_door_link_bottom"
134 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity variables reaction_force 0 x
135 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity variables reaction_force 1 y
136 solid_mechanics_definitions probes "door_link_reaction_force" integrated_surface_quantity variables reaction_force 2 z
137 solid_mechanics_definitions probes "door_hinge_reaction_force" new
138 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity
139 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity use_set_from_boundary_condition "pull_door_hinge_bolts"
140 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity variables reaction_force 0 x
141 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity variables reaction_force 1 y
142 solid_mechanics_definitions probes "door_hinge_reaction_force" integrated_surface_quantity variables reaction_force 2 z
143 solid_mechanics_definitions probes "door_hinge_max_vm" new
144 solid_mechanics_definitions probes door_hinge_max_vm field extremum part active
145 solid_mechanics_definitions probes door_hinge_max_vm field extremum part ""
146 solid_mechanics_definitions probes door_hinge_max_vm field extremum part ""
147 solid_mechanics_definitions probes door_hinge_max_vm field extremum part "door_hinge"
148 solid_mechanics_definitions probes door_hinge_max_vm field extremum maximum active
149 solid_mechanics_definitions probes door_hinge_max_vm field extremum maximum stress active
150 solid_mechanics_definitions probes door_hinge_max_vm field extremum maximum stress von_mises
151 solid_mechanics_definitions probes door_hinge_max_vm field variables stress new
152 solid_mechanics_definitions probes door_hinge_max_vm field variables stress 0 von_mises
153 #!=PROBES
154 #=-FIELD OUTPUT
155 solid_mechanics_definitions outputs "door_link_field_results" new
156 solid_mechanics_definitions outputs "door_link_field_results" field database_name "door_link_results"
157 solid_mechanics_definitions outputs "door_link_field_results" field interval "simulation_output_interval"
158 solid_mechanics_definitions outputs "door_link_field_results" field part "door_link"
159 solid_mechanics_definitions outputs "door_link_field_results" field variables displacement 0 x
160 solid_mechanics_definitions outputs "door_link_field_results" field variables displacement 1 y
161 solid_mechanics_definitions outputs "door_link_field_results" field variables displacement 2 z
162 solid_mechanics_definitions outputs "door_link_field_results" field variables stress 0 von_mises
163 solid_mechanics_definitions outputs "door_link_field_results" field variables plastic_strain 0 equivalent
164 solid_mechanics_definitions outputs "door_hinge_field_results" new
165 solid_mechanics_definitions outputs "door_hinge_field_results" field database_name "door_hinge_results"
166 solid_mechanics_definitions outputs "door_hinge_field_results" field interval "simulation_output_interval"
167 solid_mechanics_definitions outputs "door_hinge_field_results" field part "door_hinge"
168 solid_mechanics_definitions outputs "door_hinge_field_results" field variables displacement 0 x
169 solid_mechanics_definitions outputs "door_hinge_field_results" field variables displacement 1 y
170 solid_mechanics_definitions outputs "door_hinge_field_results" field variables displacement 2 z
171 solid_mechanics_definitions outputs "door_hinge_field_results" field variables stress 0 von_mises
172 solid_mechanics_definitions outputs "door_hinge_field_results" field variables plastic_strain 0 equivalent
173 solid_mechanics_definitions outputs "door_pin_field_results" new
174 solid_mechanics_definitions outputs "door_pin_field_results" field database_name "door_pin_results"
175 solid_mechanics_definitions outputs "door_pin_field_results" field interval "simulation_output_interval"
176 solid_mechanics_definitions outputs "door_pin_field_results" field part "door_pin"
177 solid_mechanics_definitions outputs "door_pin_field_results" field variables displacement 0 x
178 solid_mechanics_definitions outputs "door_pin_field_results" field variables displacement 1 y
179 solid_mechanics_definitions outputs "door_pin_field_results" field variables displacement 2 z
180 solid_mechanics_definitions outputs "door_pin_field_results" field variables stress 0 von_mises
181 solid_mechanics_definitions outputs "door_pin_field_results" field variables plastic_strain 0 equivalent
182 solid_mechanics_definitions outputs "door_link_reaction_force_results" new
183 solid_mechanics_definitions outputs "door_link_reaction_force_results" history interval "simulation_output_interval"
184 solid_mechanics_definitions outputs "door_hinge_reaction_force_results" new
185 solid_mechanics_definitions outputs "door_hinge_reaction_force_results" history interval "simulation_output_interval"
186 solid_mechanics_definitions outputs "door_hinge_reaction_force_results" history probe_variables 0 door_hinge_reaction_force
187 solid_mechanics_definitions outputs "door_hinge_max_vm_results" new
188 solid_mechanics_definitions outputs "door_hinge_max_vm_results" history interval "simulation_output_interval"
189 solid_mechanics_definitions outputs "door_hinge_max_vm_results" history probe_variables 0 door_hinge_max_vm
190 #!=FIELD OUTPUT
191 #=-PROCEDURES
192 label "door_hinge_failure"
193 procedures "pull_door_hinge" solid_mechanics
194 procedures "pull_door_hinge" solid_mechanics interval "simulation_interval"
195 procedures "pull_door_hinge" solid_mechanics flex_model "flex_models_1"
196 procedures "pull_door_hinge" solid_mechanics time_stepping_method "implicit_dynamics_quasistatic"
197 procedures "pull_door_hinge" solid_mechanics boundary_conditions 0 "hold_door_link_bottom"
198 procedures "pull_door_hinge" solid_mechanics boundary_conditions 1 "pull_door_hinge_bolts"
199 procedures "pull_door_hinge" solid_mechanics interactions 0 "frictional_contact"
200 procedures "pull_door_hinge" solid_mechanics outputs 0 "door_link_field_results"
201 procedures "pull_door_hinge" solid_mechanics outputs 1 "door_hinge_field_results"
202 procedures "pull_door_hinge" solid_mechanics outputs 2 "door_pin_field_results"
203 procedures "pull_door_hinge" solid_mechanics outputs 3 "door_link_reaction_force_results"
204 procedures "pull_door_hinge" solid_mechanics outputs 4 "door_hinge_reaction_force_results"
205 procedures "pull_door_hinge" solid_mechanics outputs 5 door_hinge_max_vm_results
206 #!=PROCEDURES
207 #=-Save the Coreform Flex project
208 save "door_lock_assembly.cf"
209 #!=Save the Coreform Flex project
210 #=-SUBMIT TRIM JOB
211 # Check the trimmed mesh
212 job "job_door_hinge_trim" new
213 job "job_door_hinge_trim" trim processor_count 16
214 job "job_door_hinge_trim" trim trim_parts [door_link door_hinge door_pin]
215 job "job_door_hinge_trim" submit
216 job "job_door_hinge_trim" wait
217 #!=SUBMIT TRIM JOB

```

```

219 #=SUBMIT SIMULATION JOB
220 job "job_door_hinge_failure" new
221 job "job_door_hinge_failure" simulation processor_count 16
222 job "job_door_hinge_failure" submit
223 job "job_door_hinge_failure" wait
224 #!=SUBMIT SIMULATION JOB
225 #=VISUALIZE SIMULATION RESULTS
226 job "job_door_hinge_failure" render_iga timestep last scalars stress von_mises
227 #!=VISUALIZE SIMULATION RESULTS
228

```

## A.2.2. job\_door\_hinge\_failure.iga

```

1 {
2   "coreform_iga_version": "2025.8",
3   "data_output_file": "cf_iga_data_output_ip2.json",
4   "flex_models": [
5     {
6       "database_name": "geom",
7       "label": "flex_models_1",
8       "parts": [
9         {
10          "label": "door_hinge_instance",
11          "material": "aluminum",
12          "material_model": "neohookean_isotropic_plasticity",
13          "part": "door_hinge",
14          "quadrature_options": {
15            "optimization_lawson_hanson_solve_completion": true,
16            "optimization_method": "tet_pruning",
17            "optimization_residual_tolerance": 1.0000000000000002e-12,
18            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
19            "quadrature": "QP1",
20            "tessellation_option": "tessellation_options_1"
21          }
22        },
23        {
24          "label": "door_link_instance",
25          "material": "aluminum",
26          "material_model": "neohookean_isotropic_plasticity",
27          "part": "door_link",
28          "quadrature_options": {
29            "optimization_lawson_hanson_solve_completion": true,
30            "optimization_method": "tet_pruning",
31            "optimization_residual_tolerance": 1.0000000000000002e-12,
32            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
33            "quadrature": "QP1",
34            "tessellation_option": "tessellation_options_1"
35          }
36        },
37        {
38          "label": "door_pin_instance",
39          "material": "steel",
40          "material_model": "neohookean_isotropic_plasticity",
41          "part": "door_pin",
42          "quadrature_options": {
43            "optimization_lawson_hanson_solve_completion": true,
44            "optimization_method": "tet_pruning",
45            "optimization_residual_tolerance": 1.0000000000000002e-12,
46            "optimized_cut_quadrature_strategy": "reduce_from_uncut",
47            "quadrature": "QP1",
48            "tessellation_option": "tessellation_options_1"
49          }
50        }
51      ],
52      "small_cell_volume_ratio": 0
53    }
54  ],
55  "functions": [
56    {
57      "label": "aluminum_plastic_hardening",
58      "piecewise_linear": {
59        "abscissa": [0, 10],
60        "ordinate": [0, 100]
61      }
62    },
63    {
64      "label": "steel_plastic_hardening",
65      "piecewise_linear": {
66        "abscissa": [0, 10],
67        "ordinate": [0, 100]
68      }
69    },
70    {
71      "constant": {

```

```

72     "value": 1
73   },
74   "label": "constant_1"
75 },
76 {
77   "label": "linear_ramp",
78   "piecewise_linear": {
79     "abscissa": [0, 1],
80     "ordinate": [0, 1]
81   }
82 },
83 ],
84 "intervals": [
85   {
86     "label": "simulation_interval",
87     "start_time": 0,
88     "stop_time": 1,
89     "time_increment": 0.0005
90   },
91   {
92     "label": "simulation_output_interval",
93     "step_increment": 1,
94     "time_increment": 1,
95     "use_start_stop_from_interval": "simulation_interval"
96   }
97 ],
98 "label": "door_hinge_failure",
99 "linear_equation_solvers": [
100   {
101     "direct": {
102       "lu": {
103         "use_diagonal_scaling": false
104       }
105     },
106     "label": "direct_lu"
107   },
108   {
109     "direct": {
110       "multi_frontal": {
111         "use_diagonal_scaling": false
112       }
113     },
114     "label": "direct_multi_frontal"
115   }
116 ],
117 "materials": [
118   {
119     "label": "aluminum",
120     "mass_density": 2.6999999999999998e-9,
121     "neohookean_isotropic_plasticity": {
122       "isotropic_plasticity": {
123         "hardening_function": "aluminum_plastic_hardening",
124         "yield_stress": 322
125       },
126       "neohookean": {
127         "poissons_ratio": 0.3,
128         "youngs_modulus": 73800
129       }
130     },
131   },
132   {
133     "label": "steel",
134     "mass_density": 2.6999999999999998e-9,
135     "neohookean_isotropic_plasticity": {
136       "isotropic_plasticity": {
137         "hardening_function": "steel_plastic_hardening",
138         "yield_stress": 2000
139       },
140       "neohookean": {
141         "poissons_ratio": 0.29,
142         "youngs_modulus": 200000
143       }
144     },
145   }
146 ],
147 "nonlinear_equation_solvers": [
148   {
149     "label": "newton_raphson",
150     "newton": {
151       "linear_equation_solver": "direct_multi_frontal",
152       "maximum_iterations": 20,
153       "target_relative_residual": 1e-8,
154       "use_line_search": false
155     }
156   }
157 ],
158 "procedures": [
159   {
160     "label": "pull_door_hinge",
161     "solid_mechanics": {
162       "boundary_conditions": [
163         "hold_door_link_bottom",

```

```

164         "pull_door_hinge_bolts"
165     ],
166     "flex_model": "flex_models_1",
167     "interactions": [
168         "frictional_contact"
169     ],
170     "interval": "simulation_interval",
171     "outputs": [
172         "door_link_field_results",
173         "door_hinge_field_results",
174         "door_pin_field_results",
175         "door_link_reaction_force_results",
176         "door_hinge_reaction_force_results"
177     ],
178     "time_stepping_method": "implicit_dynamics_quasistatic"
179 }
180 ],
181 ],
182 "solid_mechanics_definitions": {
183     "boundary_conditions": [
184         {
185             "displacement": {
186                 "components": ["x", "y", "z"],
187                 "function": "constant_1",
188                 "scale_factor": 0
189             },
190             "label": "hold_door_link_bottom",
191             "penalty": {
192                 "dimensionless": true,
193                 "value": 10
194             },
195             "quadrature_options": {
196                 "optimization_lawson_hanson_solve_completion": true,
197                 "optimization_method": "tet_pruning",
198                 "optimization_residual_tolerance": 1.0000000000000002e-12,
199                 "optimized_cut_quadrature_strategy": "reduce_from_uncut",
200                 "quadrature": "QP1",
201                 "tessellation_option": "tessellation_options_1"
202             },
203             "set": "door_link_bottom_surfaces"
204         },
205         {
206             "displacement": {
207                 "components": ["x", "y", "z"],
208                 "function": "linear_ramp",
209                 "scale_factor": [0, 0, 9]
210             },
211             "label": "pull_door_hinge_bolts",
212             "penalty": {
213                 "dimensionless": true,
214                 "value": 10
215             },
216             "quadrature_options": {
217                 "optimization_lawson_hanson_solve_completion": true,
218                 "optimization_method": "tet_pruning",
219                 "optimization_residual_tolerance": 1.0000000000000002e-12,
220                 "optimized_cut_quadrature_strategy": "reduce_from_uncut",
221                 "quadrature": "QP1",
222                 "tessellation_option": "tessellation_options_1"
223             },
224             "set": "door_hinge_bolt_surfaces"
225         }
226     ],
227     "interaction_properties": [
228         {
229             "coulomb_friction": {
230                 "coefficient_with_regularization": {
231                     "friction_coefficient": 0.3,
232                     "regularization_velocity": 0.1
233                 }
234             },
235             "label": "friction"
236         }
237     ],
238     "interactions": [
239         {
240             "label": "frictional_contact",
241             "mechanical_contact": {
242                 "activation_distance": {
243                     "dimensionless": true,
244                     "value": 0.6
245                 },
246                 "activation_zone_influential_radius_factor": 1,
247                 "angle_scaling": {
248                     "negative_dot_product_lower_limit": 0,
249                     "negative_dot_product_upper_limit": 1,
250                     "ramp_function_continuity": "C0",
251                     "scaling_strategy": "signed_distance"
252                 },
253                 "approximate_tangent_iterations": 6,
254                 "base_contact": {
255                     "complement": true

```

```

256     },
257     "collision_search_filter": {
258         "angle_filter": {
259             "threshold_angle": 30
260         },
261         "area_filter": {
262             "threshold_area_ratio": 10
263         },
264         "spurious_self_contact_filter": {
265             "use": true
266         }
267     },
268     "effective_distance_smoothing": {
269         "distance_smoothing_factor": 0.125
270     },
271     "interaction_properties": "friction",
272     "penalty": {
273         "dimensionless": true,
274         "value": 10
275     },
276     "potential_cutoff_distance_factor": 0.5
277 },
278 "quadrature_options": {
279     "optimization_lawson_hanson_solve_completion": true,
280     "optimization_method": "tet_pruning",
281     "optimization_residual_tolerance": 1.0000000000000002e-12,
282     "optimized_cut_quadrature_strategy": "reduce_from_uncut",
283     "quadrature": "QP1",
284     "tessellation_option": "tessellation_options_1"
285 }
286 ],
287 "outputs": [
288     {
289         "field": {
290             "database_name": "door_link_results_0p6",
291             "element_variable_output_strategy": "interpolate",
292             "external_surfaces_only": false,
293             "include_integration_point_output": false,
294             "interval": "simulation_output_interval",
295             "part": "door_link",
296             "variables": {
297                 "displacement": ["x", "y", "z"],
298                 "plastic_strain": ["equivalent"],
299                 "stress": ["von_mises"]
300             }
301         },
302         "label": "door_link_field_results"
303     },
304     {
305         "field": {
306             "database_name": "door_hinge_results_0p6",
307             "element_variable_output_strategy": "interpolate",
308             "external_surfaces_only": false,
309             "include_integration_point_output": false,
310             "interval": "simulation_output_interval",
311             "part": "door_hinge",
312             "variables": {
313                 "displacement": ["x", "y", "z"],
314                 "plastic_strain": ["equivalent"],
315                 "stress": ["von_mises"]
316             }
317         },
318         "label": "door_hinge_field_results"
319     },
320     {
321         "field": {
322             "database_name": "door_pin_results_0p6",
323             "element_variable_output_strategy": "interpolate",
324             "external_surfaces_only": false,
325             "include_integration_point_output": false,
326             "interval": "simulation_output_interval",
327             "part": "door_pin",
328             "variables": {
329                 "displacement": ["x", "y", "z"],
330                 "plastic_strain": ["equivalent"],
331                 "stress": ["von_mises"]
332             }
333         },
334         "label": "door_pin_field_results"
335     },
336     {
337         "history": {
338             "interval": "simulation_output_interval",
339             "probe_variables": ["door_link_reaction_force"]
340         },
341         "label": "door_link_reaction_force_results"
342     },
343     {
344         "history": {
345             "interval": "simulation_output_interval",
346             "probe_variables": ["door_hinge_reaction_force"]
347         }
348     }
349 ]

```

```

348     },
349     "label": "door_hinge_reaction_force_results"
350 },
351 ],
352 "probes": [
353     {
354         "integrated_surface_quantity": {
355             "use_set_from_boundary_condition": "hold_door_link_bottom",
356             "variables": {
357                 "reaction_force": ["x", "y", "z"]
358             }
359         },
360         "label": "door_link_reaction_force"
361     },
362     {
363         "integrated_surface_quantity": {
364             "use_set_from_boundary_condition": "pull_door_hinge_bolts",
365             "variables": {
366                 "reaction_force": ["x", "y", "z"]
367             }
368         },
369         "label": "door_hinge_reaction_force"
370     }
371 ],
372 },
373 "tessellation_options": [
374     {
375         "label": "tessellation_options_1",
376         "meshgms": {
377             "average_triangle_edge_subdivisions": 4
378         }
379     }
380 ],
381 "time_steppers": [
382     {
383         "generalized_alpha": {
384             "adaptivity": {
385                 "decrease_factor": 0.25,
386                 "increase_factor": 1.5,
387                 "maximum_time_step": 0.1,
388                 "minimum_time_step": 0.00001
389             },
390             "nonlinear_equation_solver": "newton_raphson",
391             "predictor": "constant_velocity",
392             "spectral_radius": 0
393         },
394         "label": "implicit_dynamics_quasistatic"
395     }
396 ]
397 }

```

### A.2.3. read\_force\_vm.py

```

1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Created on Thu Aug 21 10:42:45 2025
5
6  @author: danajer
7  """
8
9  import pandas as pd
10 import matplotlib.pyplot as plt
11 import numpy as np
12 import seaborn as sns
13
14 plt.close('all')
15
16 from matplotlib import rc
17
18 def set_style_sns():
19     sns.set_context('paper')
20     sns.set(font = 'serif')
21     sns.set(font_scale = 1.1)
22
23     sns.set_style('white', {
24         'font.family': 'serif',
25         'font.serif': ['Time', 'Palatino', 'serif'],
26         'lines.markersize': 10
27     })
28
29
30 plt.rcParams.update({'font.size': 8})
31 set_style_sns()

```

```

32
33 plt.rcParams.update({'text.usetex': False})
34
35 files = ['with_probe/results_0p8/cf_iga_data_output.json',
36         'with_probe/results_1p0/cf_iga_data_output.json',
37         'with_probe/results_1p2/cf_iga_data_output.json']
38 mesh_sizes = ['0p8', '1p0', '1p2']
39 directs = ['x', 'y', 'z']
40
41 plt.figure(figsize = (10,8))
42
43 force_mag = []
44 vm_stresses = []
45 dofs = np.array([260247, 154746, 101739])
46 runtimes = []
47 for ii,file in enumerate(files):
48     data = pd.read_json(file, orient = 'index')
49     data_f = data['history']['pull_door_hinge']['door_hinge_reaction_force']
50     time = data_f['time']
51     force = np.stack([data_f['reaction_force']['x'], data_f['reaction_force']['y'], data_f['reaction_force']['z']])
52     vm_stress = data['history']['pull_door_hinge']['door_hinge_max_vm']['extremum']['stress']['von_mises']
53     vm_stresses.append(vm_stress[-1])
54     force_mag.append(np.sqrt(force[0,-1]**2 + force[1,-1]**2 + force[2,-1]**2))
55     runtimes.append(data['monitors']['pull_door_hinge']['timing']['procedure_solve'][0])
56     for jj in range(3):
57         plt.subplot(2,2,jj+1)
58         plt.plot(time, force[jj,:], label = mesh_sizes[ii])
59
60
61         if ii ==2:
62             plt.xlabel('Time, s')
63             plt.ylabel('door hinge reaction force, lbf')
64             plt.title(directs[jj])
65             plt.legend()
66             plt.grid('on')
67
68     plt.subplot(2,2,4)
69     plt.plot(time, vm_stress, label = mesh_sizes[ii])
70     if ii ==2:
71         plt.xlabel('Time, s')
72         plt.ylabel('door hinge von Mises, psi')
73         plt.title(directs[jj])
74         plt.legend()
75         plt.grid('on')
76
77
78 force_mag = np.array(force_mag)
79 vm_stresses = np.array(vm_stresses)
80 runtimes = np.array(runtimes)
81
82 plt.tight_layout()
83 plt.savefig('reaction_force_comparison.png')
84
85 plt.show()
86
87
88 plt.figure(figsize = (8,8))
89 plt.subplot(2,1,1)
90 plt.plot(dofs, force_mag, '-.', label = 'IGA')
91 plt.ylabel('Reaction Force Magnitude, lbf')
92 plt.xlabel('Number of DOFs')
93 plt.grid('on')
94 plt.legend()
95 plt.tight_layout()
96
97 plt.subplot(2,1,2)
98 plt.plot(dofs, vm_stresses, '-.', label = 'IGA')
99 plt.ylabel('Peak von Mises, psi')
100 plt.xlabel('Number of DOFs')
101 plt.grid('on')
102 plt.legend()
103 plt.tight_layout()
104 plt.savefig('door_hinge_force_vm_dofs.png')
105 plt.show()

```

## A.3. Circuit Board

### A.3.1. cb\_modal\_workflow\_journal\_separate.txt

```

1 #!python
2

```



```

69 flex.cmd('structural_modal_analysis_definitions boundary_conditions fixed components 2 z')
70
71 ## output
72
73 flex.cmd('structural_modal_analysis_definitions outputs "outputs_1" new')
74 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field database_name "cb_modal_0p1"')
75 flex.cmd('intervals "intervals_1" new')
76 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field interval "intervals_1"')
77 flex.cmd('intervals intervals_1 start_time 1')
78 flex.cmd('intervals intervals_1 step_increment 1')
79 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field variables displacement new')
80 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field variables displacement 0 all')
81 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field part "part_1" ')
82
83
84 flex.cmd('set "all_parts" new ')
85 for ii in range(len(pnames)):
86     flex.cmd('set "all_parts" add parts [{}].format(ii + 1))
87
88 flex.cmd('structural_modal_analysis_definitions outputs "outputs_2" new')
89 flex.cmd('structural_modal_analysis_definitions outputs outputs_2 field database_name "cb_all_parts_modal_0p1"')
90 flex.cmd('intervals "intervals_1" new')
91 flex.cmd('structural_modal_analysis_definitions outputs outputs_2 field interval "intervals_1"')
92 flex.cmd('intervals intervals_1 start_time 1')
93 flex.cmd('intervals intervals_1 step_increment 1')
94 flex.cmd('structural_modal_analysis_definitions outputs outputs_2 field variables displacement new')
95 flex.cmd('structural_modal_analysis_definitions outputs outputs_2 field variables displacement 0 all')
96 flex.cmd('structural_modal_analysis_definitions outputs outputs_2 field set "all_parts" ')
97
98 ## functions
99
100 flex.cmd('functions "constant_1" new ')
101 flex.cmd('functions "constant_1" constant value 1 ')
102
103 ## procedure
104
105 flex.cmd('procedures procedures_1 structural_modal_analysis flex_model "flex_models_1" ')
106 flex.cmd('procedures procedures_1 structural_modal_analysis modal_solution_method "modal_solution_methods_1"')
107 flex.cmd('procedures procedures_1 structural_modal_analysis boundary_conditions new')
108 flex.cmd('procedures procedures_1 structural_modal_analysis boundary_conditions 0 fixed')
109 flex.cmd('procedures procedures_1 structural_modal_analysis interactions new')
110 flex.cmd('procedures procedures_1 structural_modal_analysis interactions 0 delete')
111 flex.cmd('procedures procedures_1 structural_modal_analysis outputs new')
112 flex.cmd('procedures procedures_1 structural_modal_analysis outputs 0 outputs_1')
113 flex.cmd('procedures procedures_1 structural_modal_analysis outputs new')
114 flex.cmd('procedures procedures_1 structural_modal_analysis outputs 1 outputs_2')
115
116 ## interactions
117
118 flex.cmd('procedures procedures_1 structural_modal_analysis interactions new')
119 flex.cmd('structural_modal_analysis_definitions interactions "interactions_1" new')
120 flex.cmd('procedures procedures_1 structural_modal_analysis interactions 0 interactions_1')
121 flex.cmd('structural_modal_analysis_definitions interactions interactions_1 tied_constraint active')
122 flex.cmd('structural_modal_analysis_definitions interaction_search_parameters "interaction_search_parameters_1" new')
123 flex.cmd('structural_modal_analysis_definitions interactions interactions_1 tied_constraint region_search_method "interaction_search_parameters_1"')
124 flex.cmd('structural_modal_analysis_definitions interactions interactions_1 tied_constraint target_set "all_parts" ')
125 flex.cmd('structural_modal_analysis_definitions interactions interactions_1 tied_constraint source_set "all_parts" ')
126
127 ## Check the trimmed mesh
128 flex.cmd('job "job_cb_modal_001" new ')
129 flex.cmd('job "job_cb_modal_001" trim processor_count 16 ')
130 flex.cmd('model_tree "job_manager queues local working_dir" "jobs_cb_modal_001" undefined')
131
132 fstr = '['
133 for ii in range(len(names)):
134     fstr = fstr + '{ } '.format(pnames[ii])
135
136 fstr = fstr + ']'
137
138 flex.cmd('job "job_cb_modal_001" trim trim_parts { } '.format(fstr))
139
140 flex.cmd('model_tree "job_manager jobs 0 job_type trim processor_count" "64" undefined')
141 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts" new')
142 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts 0" "part_1" undefined')
143 flex.cmd('model_tree "job_manager jobs 0 job_type simulation processor_count" "64" undefined')
144 flex.cmd('model_tree "job_manager jobs 0 job_type simulation render variable displacement" "undefined" undefined')
145 flex.cmd('model_tree "job_manager jobs 0 job_type simulation render variable displacement" "all" undefined')
146
147 ## end
148
149 ## intervals
150
151 flex.cmd('intervals "simulation_interval" new')
152 flex.cmd('intervals "simulation_interval" start_time 0')
153 flex.cmd('intervals intervals_1 start_step 1')
154 flex.cmd('intervals "simulation_interval" stop_time 1')
155 flex.cmd('intervals "simulation_interval" time_increment 5e-4')
156 flex.cmd('intervals "simulation_output_interval" new')
157 flex.cmd('intervals "simulation_output_interval" use_start_stop_from_interval "simulation_interval"')
158 flex.cmd('intervals "simulation_output_interval" step_increment 1')
159 #flex.cmd('intervals simulation_output_interval number_equally_spaced_values 10')
160

```

```

161 ## linear eq solvers
162
163 flex.cmd('linear_equation_solvers "direct_lu" new')
164 flex.cmd('linear_equation_solvers "direct_lu" direct_lu')
165 flex.cmd('linear_equation_solvers "direct_multi_frontal" new')
166 flex.cmd('linear_equation_solvers "direct_multi_frontal" direct_multi_frontal')
167
168 ## nonlinear eq. solvers
169
170 flex.cmd('nonlinear_equation_solvers "newton_raphson" new')
171 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton linear_equation_solver "direct_multi_frontal"')
172 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton maximum_iterations 20')
173 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton target_relative_residual 1e-8')
174
175 ## timesteppers
176
177 flex.cmd('time_steppers "implicit_dynamics_quasistatic" new ')
178 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha spectral_radius 0 ')
179 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha predictor constant_velocity ')
180 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha nonlinear_equation_solver "newton_raphson" ')
181 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity maximum_time_step 0.1 ')
182 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity minimum_time_step 1e-5 ')
183 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity decrease_factor 0.25 ')
184 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity increase_factor 1.5 ')
185
186 ## functions
187
188 flex.cmd('functions "constant_1" new ')
189 flex.cmd('functions "constant_1" constant value 1 ')
190 flex.cmd('functions "linear_ramp" new ')
191 flex.cmd('functions "linear_ramp" piecewise_linear abscissa [0 1] ')
192 flex.cmd('functions "linear_ramp" piecewise_linear ordinate [0 1] ')
193
194 ## boundary conditions
195
196 flex.cmd('set "set_bc" pick curve 5448')
197 flex.cmd('set "set_bc" pick curve 5434')
198
199 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" new ')
200 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" set "set_bc" ')
201 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement function "constant_1" ')
202 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 0 x ')
203 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 1 y ')
204 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 2 z ')
205 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement scale_factor 0 ')
206
207 ## load
208
209 flex.cmd('solid_mechanics_definitions load_conditions "load_conditions_1" new')
210 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force active ')
211 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force components new ')
212 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force components 0 y ')
213 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force function "linear_ramp" ')
214 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 part "part_1" ')
215 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force scale_factor [50] ')
216
217 ## output
218
219 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" new ')
220 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field database_name "cb_statics_result"')
221 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field interval "simulation_output_interval"')
222 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field part "part_1" ')
223 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 0 x ')
224 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 1 y ')
225 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 2 z ')
226 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables stress 0 von_mises ')
227
228 ## procedure
229
230 flex.cmd('label "uur_mech_analysis" ')
231 flex.cmd('procedures "cb_statics" solid_mechanics ')
232 flex.cmd('procedures "cb_statics" solid_mechanics interval "simulation_interval" ')
233 flex.cmd('procedures "cb_statics" solid_mechanics flex_model "flex_models_1" ')
234 flex.cmd('procedures "cb_statics" solid_mechanics time_stepping_method "implicit_dynamics_quasistatic" ')
235 flex.cmd('procedures "cb_statics" solid_mechanics boundary_conditions 0 "fixed" ')
236
237 flex.cmd('procedures "cb_statics" solid_mechanics outputs 0 "cb_statics_results" ')
238 flex.cmd('procedures "cb_statics" solid_mechanics load_conditions 0 load_conditions_1 ')
239
240 flex.cmd('model_tree "job_manager jobs 0 label" "cb_static_" undefined')
241 flex.cmd('model_tree "job_manager jobs 0 label" "cb_static_0p1" undefined')
242 flex.cmd('model_tree "job_manager jobs 0 job_type trim" "undefined" undefined')
243 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts" new')
244 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts 0" "part_1" undefined')
245 flex.cmd('model_tree "job_manager jobs 0 job_type trim processor_count" "24" undefined')

```

### A.3.2. cb\_modal\_workflow\_journal.txt

```
1 #!python
2
3 import pickle
4
5 flex.cmd('reset')
6 flex.cmd('cd "C:\\Users\\danajer\\Documents\\fy25_coreform_eval\\iga\\circuit_board" ')
7
8 flex.cmd('open "C:\\Users\\danajer\\Documents\\fy25_coreform_eval\\iga\\circuit_board\\united_cb.cf" ')
9
10 flex.cmd('materials "materials_1" new')
11 flex.cmd('materials materials_1 label "aluminum" ')
12 flex.cmd('materials aluminum mass_density 0.0002538597037 ')
13 flex.cmd('materials aluminum elastic youngs_modulus 9.9e6')
14 flex.cmd('materials aluminum elastic poissons_ratio 0.33')
15 flex.cmd('materials aluminum elastic thermal_expansion thermal_expansion_coefficient 1e-6 ')
16 flex.cmd('materials aluminum elastic thermal_expansion reference_temperature 0.0 ') # assume C
17
18 names = ['ARDUINO_UNO_REV3_SMD_NAU01_1001-003-02000_1001-003-02000_Chamfer3']
19 char_lens = [0.3]
20
21 flex.cmd('data_output_file "cf_iga_data_output_0p1.json"')
22 flex.cmd('label "cb_modal_0p1"')
23
24 pnames = []
25 for ii in range(len(names)):
26     pname = "part_{}".format(str(ii + 1))
27     flex.cmd('name part {} "{}" '.format(ii + 1, pname))
28     mesh_name = "mesh_" + pname #names[ii]
29     ms = char_lens[ii]/3.0
30     print('mesh "{}" rectilinear degree 2 continuity 1 element_size [{} {} {}] padding [2 2 2]'.format(mesh_name, ms, ms, ms))
31     flex.cmd('mesh "{}" rectilinear degree 2 continuity 1 element_size [{} {} {}] padding [2 2 2]'.format(mesh_name, ms, ms, ms))
32     flex.cmd('part {}" mesh "{}" '.format(pname, mesh_name))
33     flex.cmd('part {}" volume_box'.format(pname))
34     flex.cmd('visibility mesh [{}] toggle '.format(mesh_name))
35     pnames.append(pname)
36
37 for ii in range(len(names)):
38     flex.cmd('model_tree "geometry frames {} label" "frame-{}" undefined '.format(str(ii), str(ii)))
39
40 ## flex models
41
42 flex.cmd('flex_models "flex_models_1" new ')
43 flex.cmd('flex_models "flex_models_1" database_name "geom" ')
44 flex.cmd('flex_models "flex_models_1" small_cell_volume_ratio 0')
45
46 for ii in range(len(pnames)):
47     instance_name = "instance_" + pnames[ii]
48     flex.cmd('flex_models "flex_models_1" parts "{}" part "{}" '.format(instance_name, pnames[ii]))
49     flex.cmd('flex_models "flex_models_1" parts "{}" material "aluminum" '.format(instance_name))
50     flex.cmd('flex_models "flex_models_1" parts "{}" material_model elastic '.format(instance_name))
51
52 ## modal
53 flex.cmd('structural_modal_analysis_definitions modal_solution_methods "modal_solution_methods_1" new')
54 flex.cmd('structural_modal_analysis_definitions modal_solution_methods modal_solution_methods_1 frequency_extraction minimum_frequency 1')
55 flex.cmd('structural_modal_analysis_definitions modal_solution_methods modal_solution_methods_1 frequency_extraction maximum_frequency 5000')
56
57 ## boundary conditions
58
59 flex.cmd('set "set_bc" pick curve 5448')
60 flex.cmd('set "set_bc" pick curve 5434 ')
61
62 flex.cmd('structural_modal_analysis_definitions boundary_conditions "fixed" new ')
63 flex.cmd('structural_modal_analysis_definitions boundary_conditions "fixed" set "set_bc" ')
64 flex.cmd('structural_modal_analysis_definitions boundary_conditions fixed components new')
65 flex.cmd('structural_modal_analysis_definitions boundary_conditions fixed components 0 x')
66 flex.cmd('structural_modal_analysis_definitions boundary_conditions fixed components new')
67 flex.cmd('structural_modal_analysis_definitions boundary_conditions fixed components 1 y')
68 flex.cmd('structural_modal_analysis_definitions boundary_conditions fixed components new')
69 flex.cmd('structural_modal_analysis_definitions boundary_conditions fixed components 2 z')
70
71 ## output
72
73 flex.cmd('structural_modal_analysis_definitions outputs "outputs_1" new')
74 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field database_name "cb_modal_0p1"')
75 flex.cmd('intervals "intervals_1" new')
76 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field interval "intervals_1"')
77 flex.cmd('intervals intervals_1 start_time 1')
78 flex.cmd('intervals intervals_1 step_increment 1')
79 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field variables displacement new')
80 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field variables displacement 0 all')
81 flex.cmd('structural_modal_analysis_definitions outputs outputs_1 field part "part_1" ')
82
83 ## functions
84
85 flex.cmd('functions "constant_1" new ')
86 flex.cmd('functions "constant_1" constant value 1 ')
87
```

```

88  ## procedure
89
90  flex.cmd('procedures procedures_1 structural_modal_analysis flex_model "flex_models_1" ')
91  flex.cmd('procedures procedures_1 structural_modal_analysis modal_solution_method "modal_solution_methods_1"')
92  flex.cmd('procedures procedures_1 structural_modal_analysis boundary_conditions new')
93  flex.cmd('procedures procedures_1 structural_modal_analysis boundary_conditions 0 fixed')
94  flex.cmd('procedures procedures_1 structural_modal_analysis interactions new')
95  flex.cmd('procedures procedures_1 structural_modal_analysis interactions 0 delete')
96  flex.cmd('procedures procedures_1 structural_modal_analysis outputs new')
97  flex.cmd('procedures procedures_1 structural_modal_analysis outputs 0 outputs_1')
98
99  flex.cmd('model_tree "job_manager jobs 0 job_type trim processor_count" "64" undefined')
100 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts" new')
101 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts 0" "part_1" undefined')
102 flex.cmd('model_tree "job_manager jobs 0 job_type simulation processor_count" "64" undefined')
103 flex.cmd('model_tree "job_manager jobs 0 job_type simulation render variable displacement" "undefined" undefined')
104 flex.cmd('model_tree "job_manager jobs 0 job_type simulation render variable displacement" "all" undefined')
105
106  ## end
107
108  ## intervals
109
110 flex.cmd('intervals "simulation_interval" new')
111 flex.cmd('intervals "simulation_interval" start_time 0')
112 flex.cmd('intervals "simulation_interval" stop_time 1')
113 flex.cmd('intervals "simulation_interval" time_increment 5e-4')
114 flex.cmd('intervals "simulation_output_interval" new')
115 flex.cmd('intervals "simulation_output_interval" use_start_stop_from_interval "simulation_interval"')
116 flex.cmd('intervals "simulation_output_interval" step_increment 1')
117 #flex.cmd('intervals simulation_output_interval number_equally_spaced_values 10')
118
119  ## linear eq solvers
120
121 flex.cmd('linear_equation_solvers "direct_lu" new')
122 flex.cmd('linear_equation_solvers "direct_lu" direct lu')
123 flex.cmd('linear_equation_solvers "direct_multi_frontal" new')
124 flex.cmd('linear_equation_solvers "direct_multi_frontal" direct multi_frontal')
125
126  ## nonlinear eq. solvers
127
128 flex.cmd('nonlinear_equation_solvers "newton_raphson" new')
129 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton linear_equation_solver "direct_multi_frontal"')
130 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton maximum_iterations 20')
131 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton target_relative_residual 1e-8')
132
133  ## timesteppers
134
135 flex.cmd('time_steppers "implicit_dynamics_quasistatic" new ')
136 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha spectral_radius 0 ')
137 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha predictor constant_velocity ')
138 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha nonlinear_equation_solver "newton_raphson" ')
139 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity maximum_time_step 0.1 ')
140 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity minimum_time_step 1e-5 ')
141 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity decrease_factor 0.25 ')
142 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity increase_factor 1.5 ')
143
144  ## functions
145
146 flex.cmd('functions "constant_1" new ')
147 flex.cmd('functions "constant_1" constant value 1 ')
148 flex.cmd('functions "linear_ramp" new ')
149 flex.cmd('functions "linear_ramp" piecewise_linear abscissa [0 1] ')
150 flex.cmd('functions "linear_ramp" piecewise_linear ordinate [0 1] ')
151
152  ## boundary conditions
153
154 flex.cmd('set "set_bc" pick curve 5448')
155 flex.cmd('set "set_bc" pick curve 5434 ')
156
157 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" new ')
158 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" set "set_bc" ')
159 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement function "constant_1" ')
160 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 0 x ')
161 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 1 y ')
162 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 2 z ')
163 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement scale_factor 0 ')
164
165  ## load
166
167 flex.cmd('solid_mechanics_definitions load_conditions "load_conditions_1" new')
168 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force active ')
169 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force components new ')
170 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force components 0 y ')
171 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force function "linear_ramp" ')
172 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 part "part_1" ')
173 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force scale_factor [50] ')
174
175  ## output
176
177 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" new ')
178 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field database_name "cb_statics_result"')
179 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field interval "simulation_output_interval"')

```

```

180 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field part "part_1" ')
181 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 0 x ')
182 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 1 y ')
183 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 2 z ')
184 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables stress 0 von_mises ')
185
186 ## procedure
187
188 flex.cmd('label "uur_mech_analysis" ')
189 flex.cmd('procedures "cb_statics" solid_mechanics ')
190 flex.cmd('procedures "cb_statics" solid_mechanics interval "simulation_interval" ')
191 flex.cmd('procedures "cb_statics" solid_mechanics flex_model "flex_models_1" ')
192 flex.cmd('procedures "cb_statics" solid_mechanics time_stepping_method "implicit_dynamics_quasistatic" ')
193 flex.cmd('procedures "cb_statics" solid_mechanics boundary_conditions 0 "fixed" ')
194
195 flex.cmd('procedures "cb_statics" solid_mechanics outputs 0 "cb_statics_results" ')
196 flex.cmd('procedures "cb_statics" solid_mechanics load_conditions 0 load_conditions_1 ')
197
198 flex.cmd('model_tree "job_manager jobs 0 label" "cb_static_" undefined')
199 flex.cmd('model_tree "job_manager jobs 0 label" "cb_static_0p1" undefined')
200 flex.cmd('model_tree "job_manager jobs 0 job_type trim" "undefined" undefined')
201 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts" new')
202 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts 0" "part_1" undefined')
203 flex.cmd('model_tree "job_manager jobs 0 job_type trim processor_count" "24" undefined')

```

### A.3.3. cb\_workflow\_journal.txt

```

1 #!python
2
3 import pickle
4
5 flex.cmd('reset')
6 flex.cmd('cd "C:\Users\danajer\Documents\fy25_coreform_eval\iga\circuit_board" ')
7
8 flex.cmd('open "C:\Users\danajer\Documents\fy25_coreform_eval\iga\circuit_board\united_cb.cf" ')
9
10 flex.cmd('materials "materials_1" new')
11 flex.cmd('materials materials_1 label "aluminum" ')
12 flex.cmd('materials aluminum mass_density 0.0002538597037 ')
13 flex.cmd('materials aluminum elastic youngs_modulus 9.9e6')
14 flex.cmd('materials aluminum elastic poissons_ratio 0.33')
15 flex.cmd('materials aluminum elastic thermal_expansion thermal_expansion_coefficient 1e-6 ')
16 flex.cmd('materials aluminum elastic thermal_expansion reference_temperature 0.0 ') # assume C
17
18 names = ['ARDUINO_UNO_REV3_SMD_NAU01_1001-003-02000_1001-003-02000_Chamfer3']
19 char_lens = [0.3]
20
21 flex.cmd('data_output_file "cf_iga_data_output_0p1.json"')
22 flex.cmd('label "cb_static_grav_0p1"')
23
24 pnames = []
25 for ii in range(len(names)):
26     pname = "part_{}".format(str(ii + 1))
27     flex.cmd(' name part {} "{}" '.format(ii + 1, pname))
28     mesh_name = "mesh_" + pname #names[ii]
29     ms = char_lens[ii]/3.0
30     print('mesh "{}" rectilinear degree 2 continuity 1 element_size [{} {} {}] padding [2 2 2]'.format(mesh_name, ms, ms, ms))
31     flex.cmd('mesh "{}" rectilinear degree 2 continuity 1 element_size [{} {} {}] padding [2 2 2]'.format(mesh_name, ms, ms, ms))
32     flex.cmd('part "{}" mesh "{}" '.format(pname, mesh_name))
33     flex.cmd('part "{}" volume_box'.format(pname))
34     flex.cmd(' visibility mesh [{}] toggle '.format(mesh_name))
35     pnames.append(pname)
36
37 for ii in range(len(names)):
38     flex.cmd('model_tree "geometry frames {} label" "frame-{}" undefined '.format(str(ii),str(ii)))
39
40 ## flex models
41
42 flex.cmd('flex_models "flex_models_1" new ')
43 flex.cmd('flex_models "flex_models_1" database_name "geom" ')
44 flex.cmd('flex_models "flex_models_1" small_cell_volume_ratio 0')
45
46 for ii in range(len(pnames)):
47     instance_name = "instance_" + pnames[ii]
48     flex.cmd('flex_models "flex_models_1" parts "{}" part "{}" '.format(instance_name, pnames[ii]))
49     flex.cmd('flex_models "flex_models_1" parts "{}" material "aluminum" '.format(instance_name))
50     flex.cmd('flex_models "flex_models_1" parts "{}" material_model elastic '.format(instance_name))
51
52 ## intervals
53
54 flex.cmd('intervals "simulation_interval" new')
55 flex.cmd('intervals "simulation_interval" start_time 0')
56 flex.cmd('intervals "simulation_interval" stop_time 1')
57 flex.cmd('intervals "simulation_interval" time_increment 5e-4')

```

```

58 flex.cmd('intervals "simulation_output_interval" new')
59 flex.cmd('intervals "simulation_output_interval" use_start_stop_from_interval "simulation_interval"')
60 #flex.cmd('intervals "simulation_output_interval" step_increment 1')
61 flex.cmd('intervals simulation_output_interval number_equally_spaced_values 10')
62
63 ## linear eq solvers
64
65 flex.cmd('linear_equation_solvers "direct_lu" new')
66 flex.cmd('linear_equation_solvers "direct_lu" direct lu')
67 flex.cmd('linear_equation_solvers "direct_multi_frontal" new')
68 flex.cmd('linear_equation_solvers "direct_multi_frontal" direct multi_frontal')
69
70 ## nonlinear eq. solvers
71
72 flex.cmd('nonlinear_equation_solvers "newton_raphson" new')
73 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton linear_equation_solver "direct_multi_frontal"')
74 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton maximum_iterations 20')
75 flex.cmd('nonlinear_equation_solvers "newton_raphson" newton target_relative_residual 1e-8')
76
77 ## timesteppers
78
79 flex.cmd('time_steppers "implicit_dynamics_quasistatic" new ')
80 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha spectral_radius 0 ')
81 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha predictor constant_velocity ')
82 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha nonlinear_equation_solver "newton_raphson" ')
83 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity maximum_time_step 0.1 ')
84 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity minimum_time_step 1e-5 ')
85 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity decrease_factor 0.25 ')
86 flex.cmd('time_steppers "implicit_dynamics_quasistatic" generalized_alpha adaptivity increase_factor 1.5 ')
87
88 ## functions
89
90 flex.cmd('functions "constant_1" new ')
91 flex.cmd('functions "constant_1" constant value 1 ')
92 flex.cmd('functions "linear_ramp" new ')
93 flex.cmd('functions "linear_ramp" piecewise_linear abscissa [0 1] ')
94 flex.cmd('functions "linear_ramp" piecewise_linear ordinate [0 1] ')
95
96 ## boundary conditions
97
98 flex.cmd('set "set_bc" pick curve 5448')
99 flex.cmd('set "set_bc" pick curve 5434')
100
101 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" new ')
102 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" set "set_bc" ')
103 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement function "constant_1" ')
104 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 0 x ')
105 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 1 y ')
106 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement components 2 z ')
107 flex.cmd('solid_mechanics_definitions boundary_conditions "fixed" displacement scale_factor 0 ')
108
109 ## load
110
111 flex.cmd('solid_mechanics_definitions load_conditions "load_conditions_1" new')
112 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force active ')
113 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force components new ')
114 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force components 0 y ')
115 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force function "linear_ramp" ')
116 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 part "part_1" ')
117 flex.cmd('solid_mechanics_definitions load_conditions load_conditions_1 body_force scale_factor [50] ')
118
119 ## output
120
121 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" new ')
122 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field database_name "cb_statics_result"')
123 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field interval "simulation_output_interval"')
124 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field part "part_1" ')
125 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 0 x ')
126 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 1 y ')
127 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables displacement 2 z ')
128 flex.cmd('solid_mechanics_definitions outputs "cb_statics_results" field variables stress 0 von_mises ')
129
130 ## procedure
131
132 flex.cmd('label "uur_mech_analysis" ')
133 flex.cmd('procedures "cb_statics" solid_mechanics ')
134 flex.cmd('procedures "cb_statics" solid_mechanics interval "simulation_interval" ')
135 flex.cmd('procedures "cb_statics" solid_mechanics flex_model "flex_models_1" ')
136 flex.cmd('procedures "cb_statics" solid_mechanics time_stepping_method "implicit_dynamics_quasistatic" ')
137 flex.cmd('procedures "cb_statics" solid_mechanics boundary_conditions 0 "fixed" ')
138
139 flex.cmd('procedures "cb_statics" solid_mechanics outputs 0 "cb_statics_results" ')
140 flex.cmd('procedures "cb_statics" solid_mechanics load_conditions 0 load_conditions_1 ')
141
142 flex.cmd('model_tree "job_manager jobs 0 label" "cb_static_" undefined')
143 flex.cmd('model_tree "job_manager jobs 0 label" "cb_static_0p1" undefined')
144 flex.cmd('model_tree "job_manager jobs 0 job_type trim" "undefined" undefined')
145 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts" new')
146 flex.cmd('model_tree "job_manager jobs 0 job_type trim parts 0" "part_1" undefined')

```

#### A.4. UUR Discriminator

DRAFT



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Laboratories

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